

BHUVAN RAJANAHALLY JAYAKUMAR

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[Portfolio](#) - [LinkedIn](#) - [GitHub](#) - **Expected Graduation:** December 2026

SUMMARY

MS Computer Science candidate and software engineer with production experience designing, building, and operating a scalable Python microservice through CI/CD, plus fault-tolerant distributed systems work and LLM multi-agent research published as a first-author preprint.

EDUCATION

Master of Science in Computer Science, University of Southern California, December 2026

B.E. in Computer Science and Engineering, Visvesvaraya Technological University, July 2023

EXPERIENCE

STYLEBOT

Software Engineering Intern

April 2025 - August 2025

- Owned the full lifecycle of a scalable Python/FastAPI NLP microservice, from design through deployment, serving 20+ enterprise media clients at 50K+ requests/day (peak 10K/hour)
- Cut p95 response latency 52% (2.3s to 1.1s) on the microservice by adding a Redis caching layer with a one-day TTL and async database access to eliminate blocking I/O
- Shipped production updates through a Bitbucket-Heroku CI/CD pipeline with 93% test coverage on the engine, performing code reviews to keep releases reliable

INDIAN INSTITUTE OF SCIENCE

AI Research Intern

October 2023 - May 2024

- Fine-tuned a YOLO-pose model for medical human-pose detection, lifting keypoint accuracy (PCK) from 70% to 95% through iterative training and targeted error analysis
- Cut per-image annotation time 75% (60s to 15s) by building annotation tooling and data pipelines in Python, accelerating creation of a 10,000+ image keypoint dataset
- Engineered an on-device inference pipeline running the YOLO-pose model at 20 FPS with 90%+ accuracy on a consumer laptop GPU, enabling a real-time post-surgery physical therapy companion

PROJECTS

- [Preference Falsification in LLM Multi-Agent Networks](#)**: Built a dual-channel multi-agent simulation with Google ADK and LangChain on Vertex AI, running up to 12 concurrent agents through multi-round deliberation to measure gaps between private beliefs and public statements under social pressure; first-author preprint
- [Project Hydra \(Distributed Systems Engineering Platform\)](#)**: Built a scalable, fault-tolerant distributed systems platform running Kafka-backed microservices across 8 to 30 Docker containers, driven by a k6 load-test harness at roughly 1K to 10K RPS; fits a Universal Scalability Law model to live measurements to project saturation points, surfacing cache stampedes, replica lag, and hot-partition contention
- [Auto-Apply \(Python, FastAPI, pytest\)](#)**: Cut time per job application from 5 minutes to 45 seconds by building a Python/FastAPI and Chrome Extension tool that tailors one-page LaTeX resumes and auto-fills forms with browser-use; used for 100+ applications in a week

TECHNICAL SKILLS

Programming Languages: Java, Python, C/C++, JavaScript, SQL

Frameworks: FastAPI, Flask, Node.js, LangChain, Google ADK

Databases & Messaging: PostgreSQL, MySQL (SQL), MongoDB (NoSQL), Redis, Kafka

Cloud & DevOps: AWS (S3, EC2, Lambda), Docker, CI/CD, GitHub Actions, GCP (Vertex AI)

AI / ML: AI-Assisted Development (Claude Code, Cursor), Model Context Protocol, AI Agent Tooling, LLM Application Development, LLM Multi-Agent Systems, ONNX Runtime Web, NLP, PyTorch, Computer Vision

Tools & Platforms: Git, PyTest, Playwright, Unix/Linux, gRPC, WebAssembly, Web Workers, k6, browser-use

Concepts: Microservices, Distributed and Parallel Systems, System Design, REST/JSON APIs, Agile/SDLC

Coursework: Analysis of Algorithms, Database Systems, Applied NLP

ACHIEVEMENTS

- Early Career Innovator Award, University of Southern California, 2025
- Google Summer of Code, 2022